

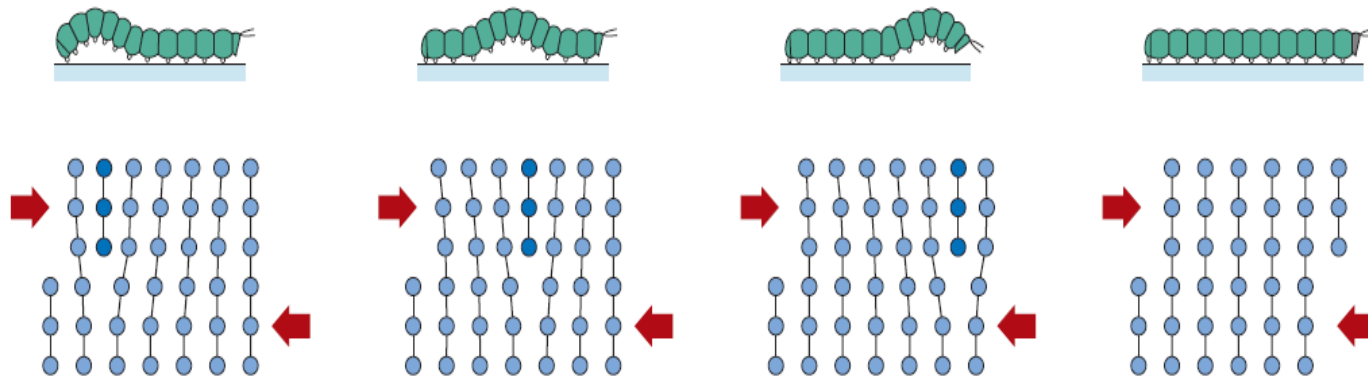
TOPIC 5. MECHANICAL PROPERTIES II

- Strengthening Mechanisms:
 - Strain Hardening
 - Solid Solution
 - Grain Size Reduction
 - Precipitation Hardening
- Hardness:
 - Definition, qualitative and quantitative testing
 - Brinell, Rockwell and Vickers
 - Relation of hardness with other mechanical properties

STRENGTHENING MECHANISMS

Dislocations and plastic deformation

- **Plastic deformation under shear stress = movement of dislocations (many)!!!**
- Dislocations move along a **slip plane** in a **slip direction**
- **Breaking and formation of bonds** in dislocation movement is **sequential**
- The process by which plastic deformation is produced by dislocation motion is termed **slip**



The plastic deformation → dislocations movement/motion

DISLOCATION MOTION

https://www.youtube.com/watch?v=BV1cxwxnhPs&ab_channel=JoshKacher



↑ Dislocation motion $\Rightarrow$ ↓ Strength

STRENGTHENING MECHANISMS

The capability of a metal to be plastically deformed depends on the capability of dislocations to move. Therefore:

Restriction towards the movement of dislocations → ↑ stress required to activate the slip system → ↑ STRENGTHENING

Strengthening methods rely on one principle:
impeding the movement of dislocations

TYPES:

- Strain Hardening
- Solid Solution Strengthening
- Grain Size Reduction Strengthening
- Precipitation Strengthening

STRAIN HARDENING

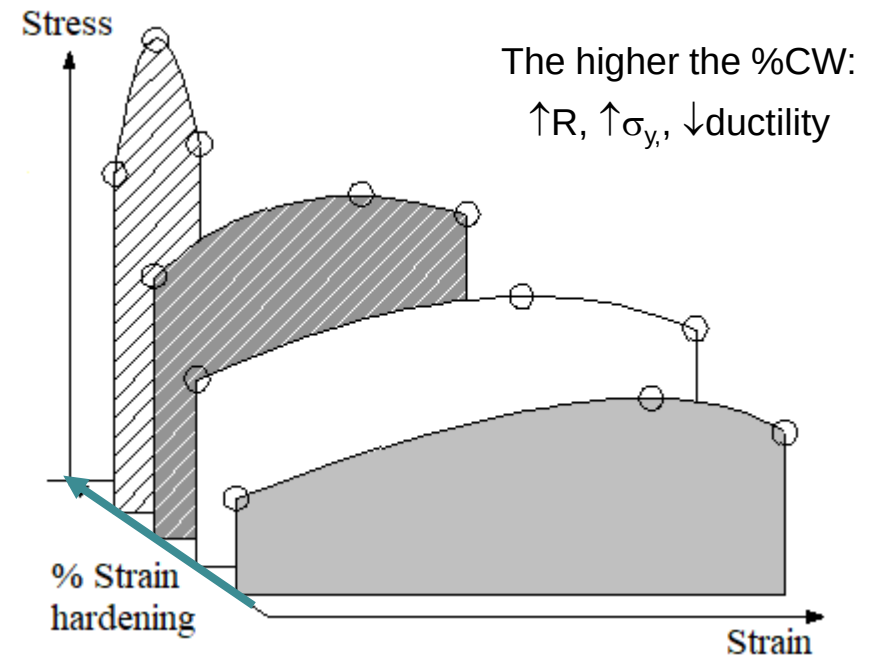
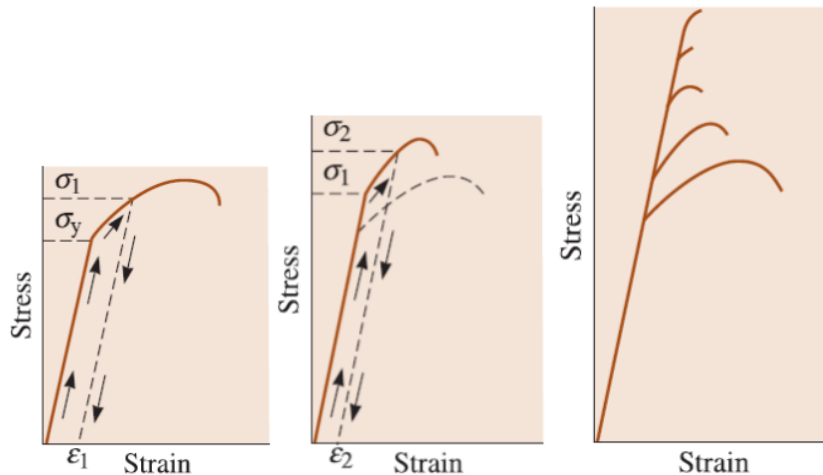
“Tangle the dislocations with each other”

Increase in strength and hardness when a material is plastically deformed (at low temperatures).
Also called cold work

The cold plastic deformation generates dislocations (from 10^6 m/m^3 to 10^{12} m/m^3) and the increase in dislocation density makes it difficult for further plastic deformation

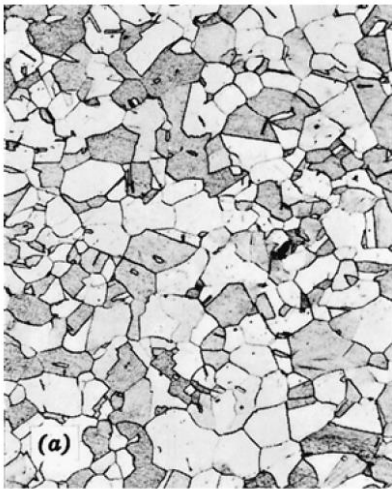
⇒ Dislocations **entangle** with one another ⇒ ↓ Dislocation motion ⇒ ↑ Strengthening

Degree of plastic deformation ↔ % cold work (%CW)



STRAIN HARDENING

⇒ If it is considered in the process **design** ⇒
during processing the strength and hardness of
the material ↑ ⇒ material cost ↓



Cold work produces texture:

“Regular orientation of the grains in the direction of rolling”.



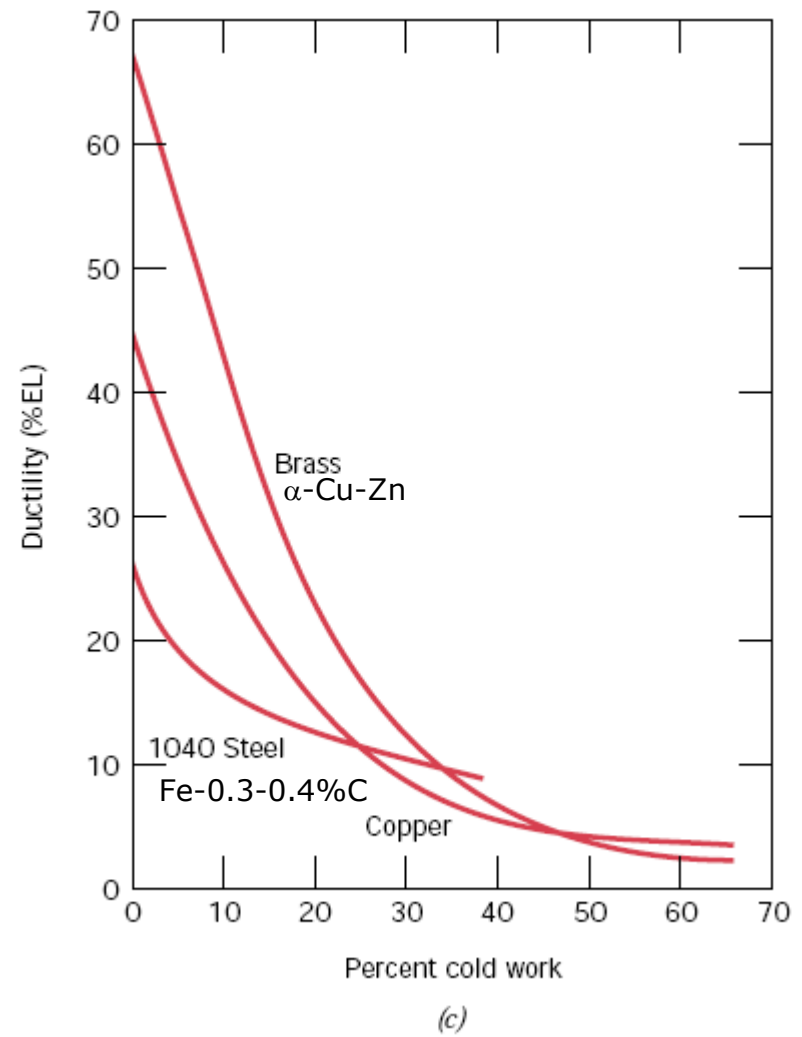
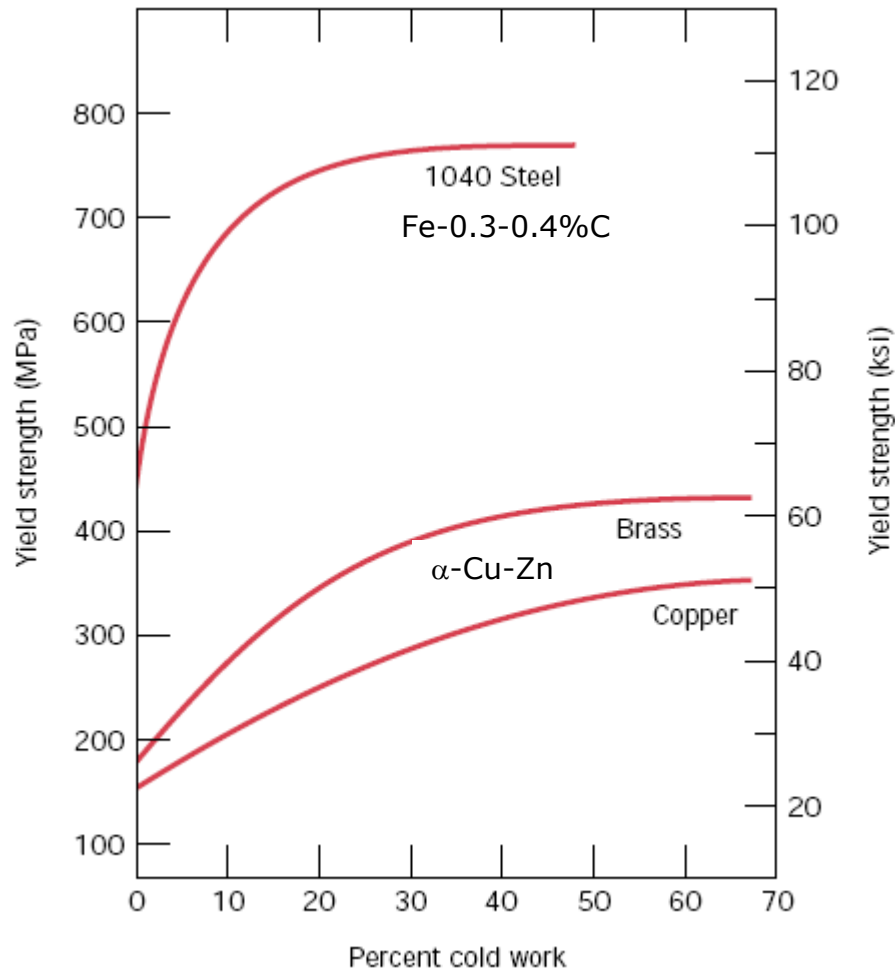
Degree of plastic deformation
<> **% cold work (%CW)**

$$\%CW = \left(\frac{A_0 - A_d}{A_0} \right) \cdot 100$$

A_0 : original cross-section area

A_d : cross-section area after deformation

STRAIN HARDENING



(c)

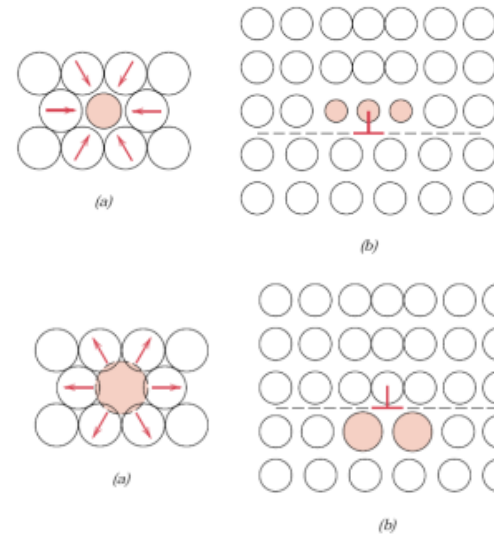
SOLID SOLUTION STRENGTHENING

“Make an irregular pattern in the lattice”

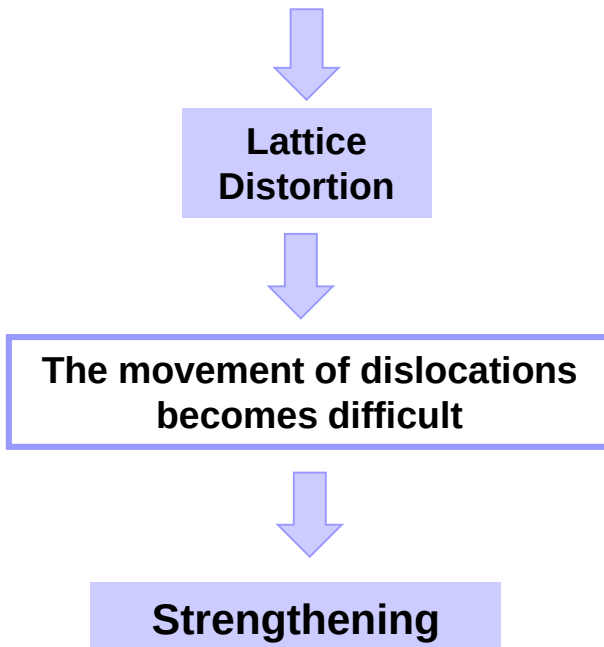
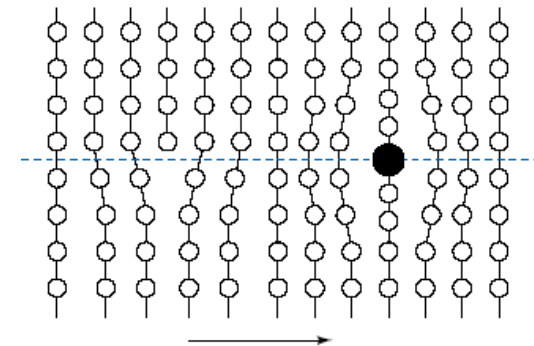
Increase of the strength of the material by introducing impurities whether as substitutional or interstitial atoms (e.g. by forming solid solutions) = **alloying**

A vacancy occupied by a solute of **smaller radius**:
atoms of the solvent submitted to **tension**

A vacancy occupied by a solute of **greater radius**:
atoms of the solvent submitted to **compression**



Possible location of solute atoms regarding a dislocation to cancel some of the strain in the lattice surrounding a dislocation



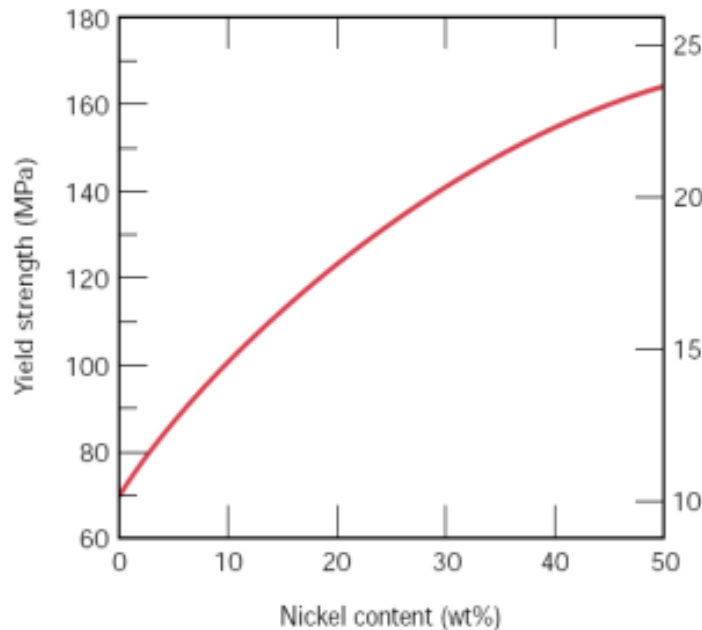
SOLID SOLUTION STRENGTHENING

Alloying elements:

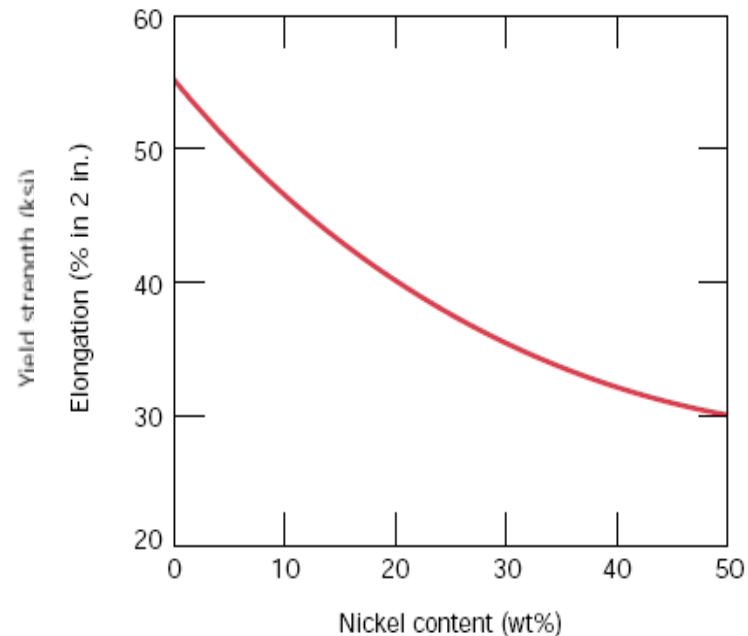
- Improve mechanical properties (yield strength, UTS, hardness)
- Decrease ductility, $R_{\text{corrosion}}$, σ_{electric}

Alloys are stronger than pure metals because impurity atoms impose strains on the surrounding host atoms that restrict dislocation movement.

Cu-Ni



(b)



(c)



If solubility limit is exceeded

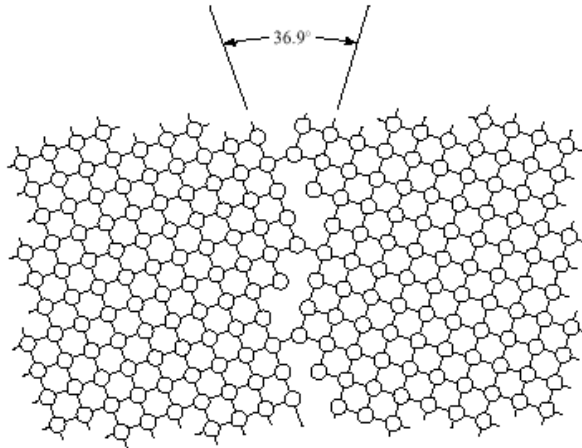
PRECIPITATION HARDENING

GRAIN SIZE REDUCTION STRENGTHENING

“Limit the space to move”

The grain boundaries separate crystals with different orientations.

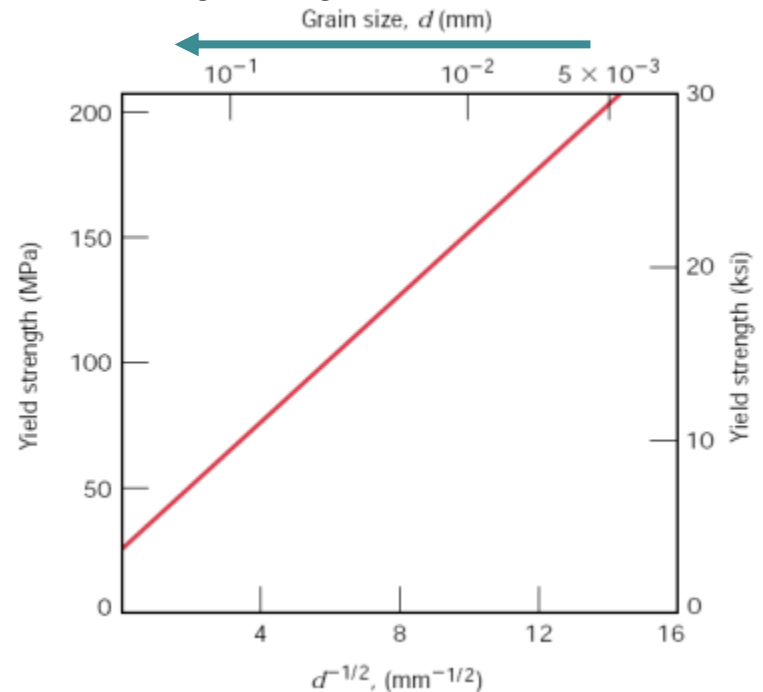
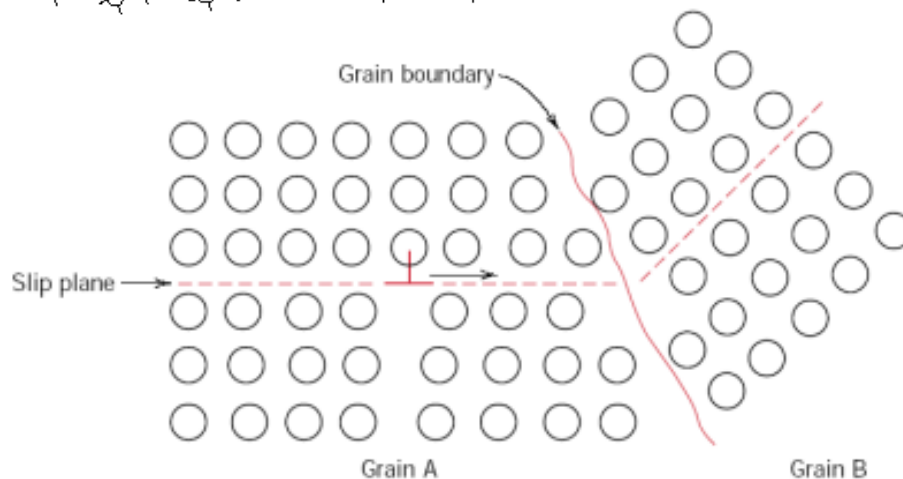
The grain boundary width: 2-5 interatomic spaces



The grain boundary makes difficult dislocations sliding →
Angle between grains

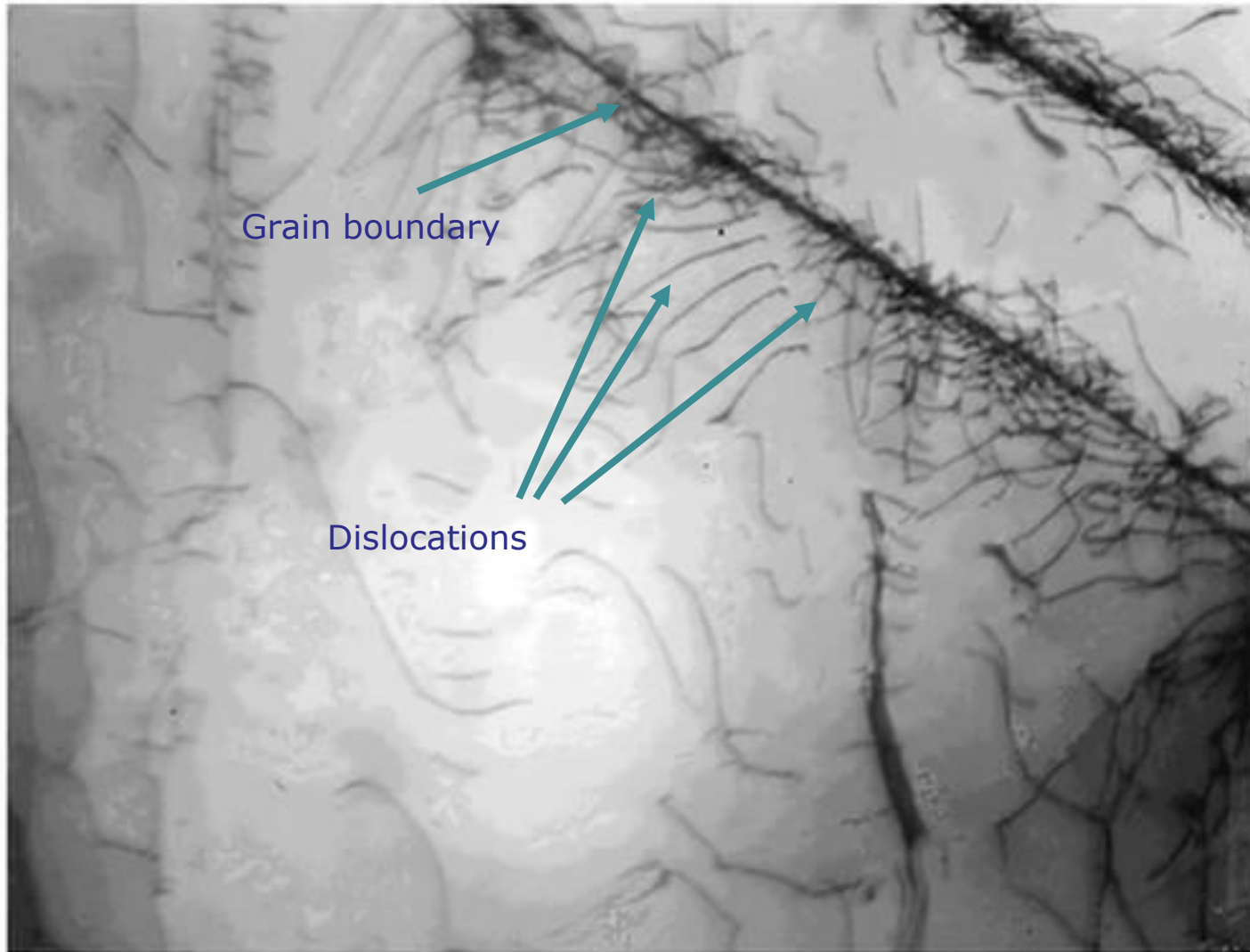
Grains act as barriers to slip:

↓ Grain size ⇒ ↑ grains ⇒ ↑ **barriers** to dislocation motion ⇒ ⇒ ↑ strengthening



GRAIN SIZE REDUCTION STRENGTHENING

Grain boundaries block the dislocations

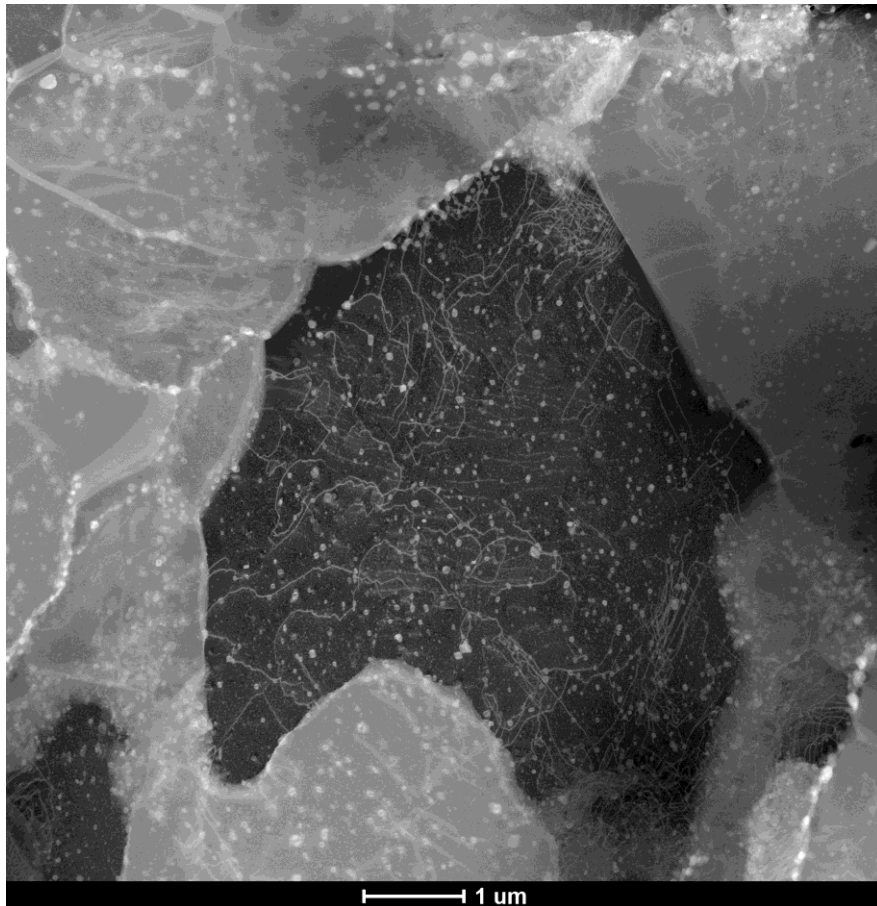


PRECIPITATION STRENGTHENING

“Block dislocations using foreign substances”

Particles of a secondary phase precipitate in the alloy, which act as obstacles for dislocations motion

The precipitates distort the crystalline structure of the matrix



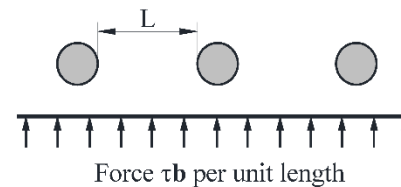
Precipitates:

↓ size, ↑ dispersion, ↑ density

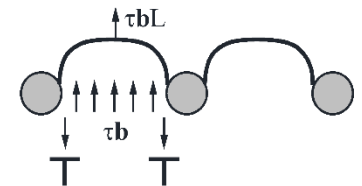


↑ strengthening

a) Approach situation



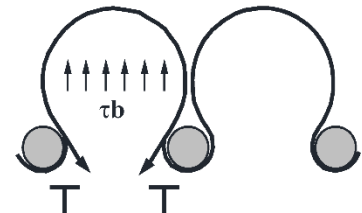
c) Critical situation



b) Sub-critical situation

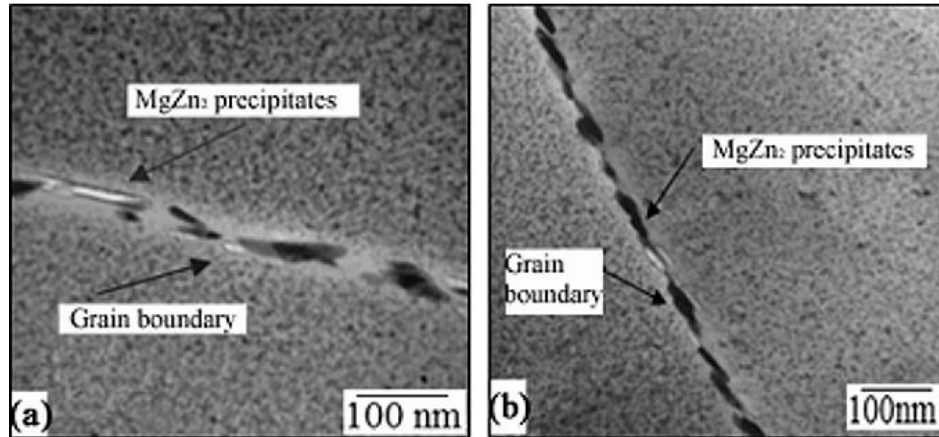


c) Escape situation



PRECIPITATION STRENGTHENING

Application: Aluminum alloys for aircraft applications (Series 2xxx, 7xxx)



Material	UTS (MPa)
Pure Al	45
Al alloy (SS strengthening)	110
Al alloy (Cold worked)	165
Al-Cu-Mg alloy (Prec. Strengt.)	469
Al-Cu-Mg-Zn alloy(Prec. Strengt.)	572

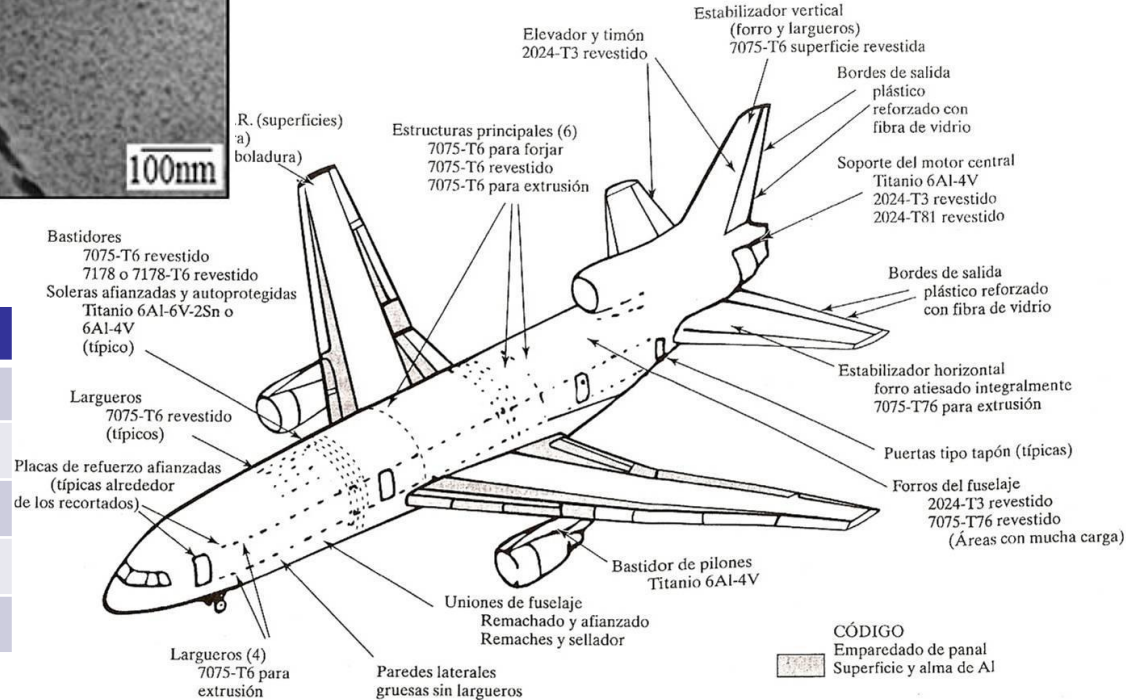


Fig. 13-1 Uso de aleaciones de aluminio 7XXX y 2XXX en un avión comercial [3].

TOPIC 5. MECHANICAL PROPERTIES II

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HARDNESS

Hardness: Surface resistance to permanent (plastic) deformation

How is it measured?

Measure resistance to abrasion

Mohs
Martens
Wear test

Measure resistance to indentation.
Measure the size /depth of a indentation
performed with a specific load

Brinell
Vickers
Rockwell

These types of tests are used very much in industry because :

- They are fast, simple and inexpensive
- They are non destructive

The way you measure it is the way you define it!

No universal conversion between results of these methods.

HARDNESS

Abrasion resistance

1. Mohs Scale (1820)

Applications in geology and mineralogy

Every material can be scratched by the materials that are placed further up in the scale

Moh's hardness kit



Finger nail ($H = 2.5$)



Razor blade ($H = 5.5$)

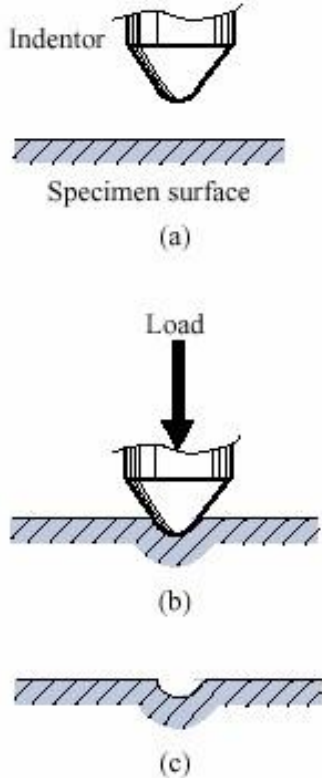
Knife ($H = 6,5$)

Material	Scale
Talc	1
Plaster	2
Limestone	3
Fluorite	4
Apatite	5
Feldspar	6
Quartz	7
Topaz	8
Corundum	9
Diamond	10

HARDNESS

Resistance to indentation (localized plastic deformation)

Measure of indentation performed with an indenter



Classification = f (load and indentation)

- Brinell Test (HBW)
- Vickers Test (HV)
- Rockwell Test (HR)

Test of **surface hardness** (coatings):

- Vickers and Rockwell

Test of **microhardness**:

- Vickers, Knoop.

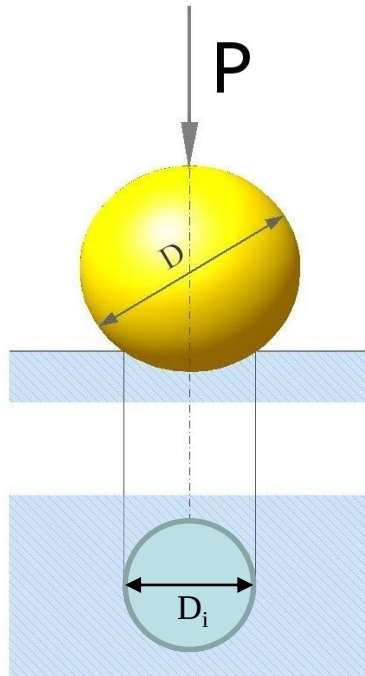


BRINELL HARDNESS (HB)

BRINELL HARDNESS (HB)

Indenter: Ball of WC or hardened steel, diameter **D**

Nomenclature: HBW or HB or BHN



$$HB = \frac{P}{S} = \frac{P}{\pi \frac{D}{2} (D - \sqrt{D^2 - D_i^2})}$$

$$HBW = K_{te} \cdot \frac{\text{Force}}{\text{Dent area}}$$

P= Applied weight in kg

S= Surface of spherical casqued in mm²

D= Diameter of the ball in mm (WC ball)

D_i= Diameter of spherical casqued.

BRINELL HARDNESS (HB)

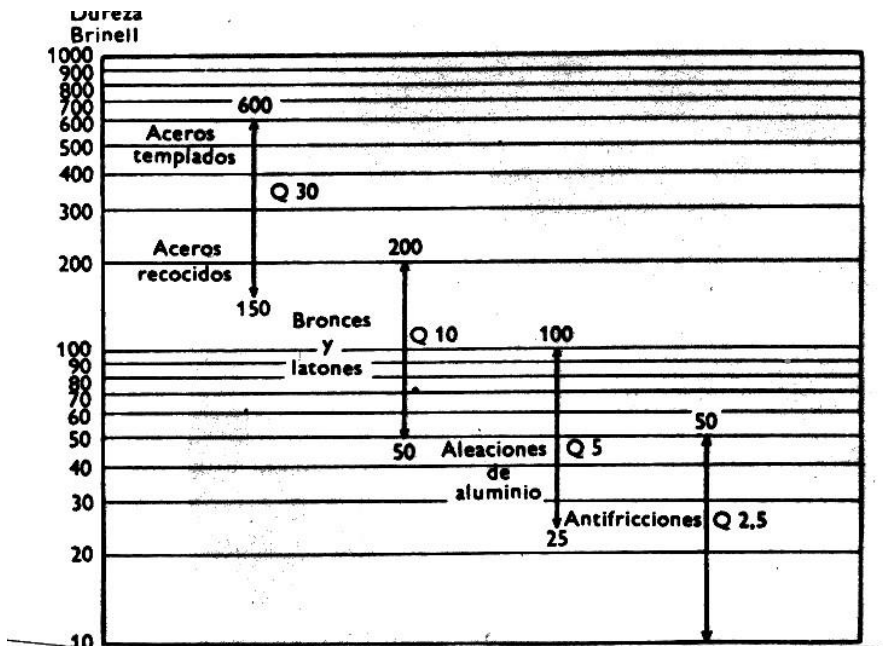
Load to be applied:

$$P = QD^2 = cte$$

P = test load

D= Ø of the ball

$$\text{Load factor} = Q = \frac{P}{D^2} = cte$$



Ball diameter D mm	Load factor Q			
	30	10	5	2,5
	Load in kg			
10	3000	1000	500	250
5	750	250	125	62,5
2,5	187,5	62,5	31,25	15,6
1,25	46,9	15,6	7,81	3,91
0,625	11,7	3,91	1,953	0,977

For small thickness $\Rightarrow D \downarrow$ (5, 2.5, 1.25 and 0.625 mm)

https://www.youtube.com/watch?v=BHZALtqAjeM&ab_channel=RealEngineering

ROCKWELL HARDNESS (HR)

https://www.youtube.com/watch?v=p-lu7mdT7ns&ab_channel=SIInstrumentsTV

ROCKWELL HARDNESS (HR)

The hardness is a function of the depth of penetration.

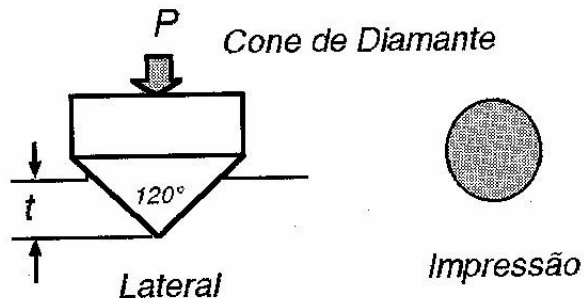
- Nomenclature: HRX
- Indenters:
 - 1) balls of hardened steel
 - 2) Diamond cone cut at an angle of 120°
- Preload: 10 or 3 kg
- Load: 65 - 150 kg

Table 6.6a Rockwell Hardness Scales

Scale Symbol	Indenter	Major Load (kg)
A	Diamond	60
B	$\frac{1}{16}$ -in. ball	100
C	Diamond	150
D	Diamond	100
E	$\frac{1}{8}$ -in. ball	100
F	$\frac{1}{16}$ -in. ball	60
G	$\frac{1}{16}$ -in. ball	150
H	$\frac{1}{8}$ -in. ball	60
K	$\frac{1}{8}$ -in. ball	150

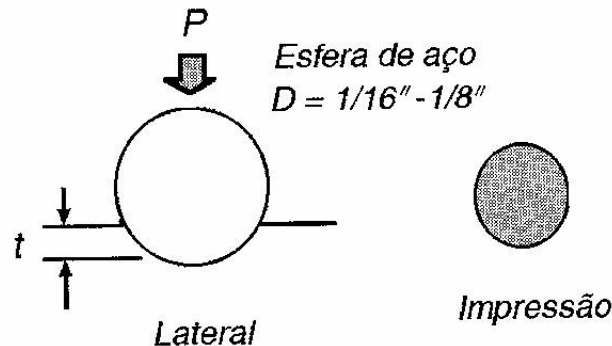
Rockwell (A, C, D)

P (60 kgf, 150 kgf, 100 kgf)



Rockwell (B, F, G)

P (100 kgf, 60 kgf, 150 kgf)



ROCKWELL HARDNESS (HR)

We apply 2 successive loads with an indenter and we measure h .

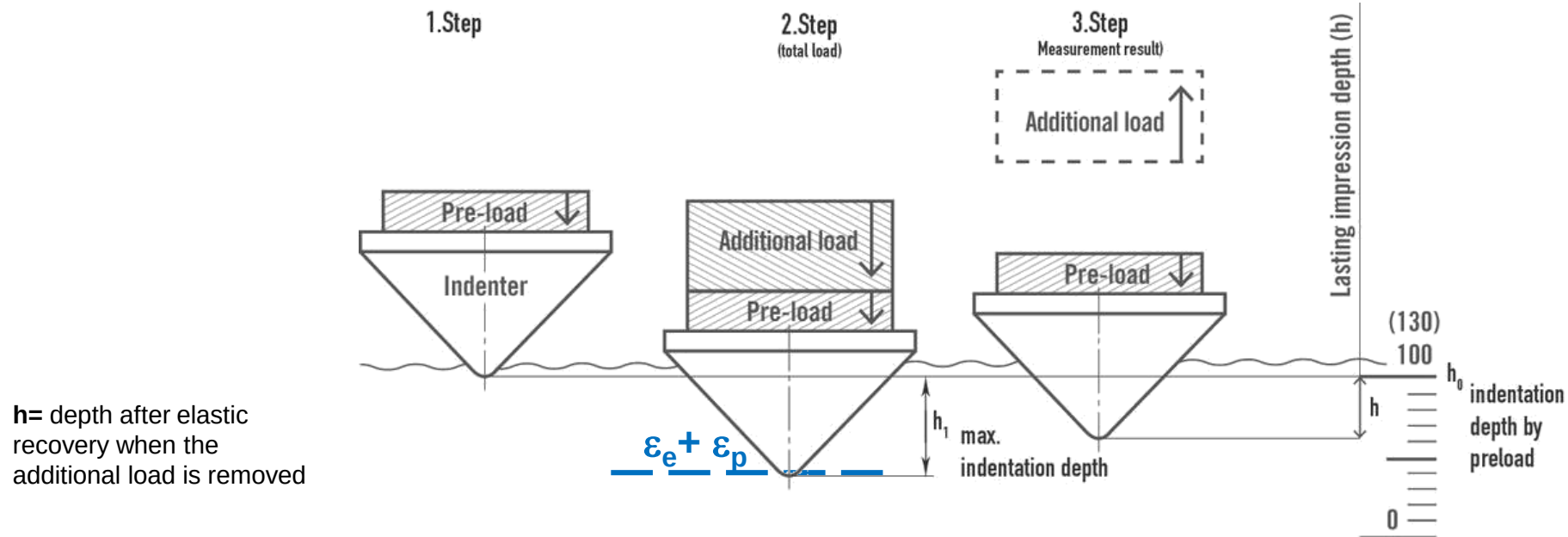
$$\text{Hardness HR} = N - \frac{h}{S}$$

N = N° specific for each scale

S = Specific unit for each scale

h = Remaining increase of the depth of penetration.

	S	N
HRA, HRC, HRD	0,002	100
HRB, HRE, HRF, HRG, HRK	0,002	130
HRT, HRN -coatings-	0,001	100



DIFFERENT SCALES AS A FUNCTION OF LOAD AND INDENTER.

Escala Designación	Tipo de prueba	Tipo y tamaño del penetrador	Carga menor en Kg.	Carga mayor en Kg.	Escala del comparador		Aplicaciones
					Color	Colocación	
A	Normal	Cono de diamante	10	60	Negro	Fuera	Aceros nitrurados, flejes estirados en frío, hojas de afeitar. Carburos metálicos (90 a 98).
B	Normal	Bola de 1/16"	10	100	Rojo	Dentro	Aceros al carbono recocidos de bajo contenido en C.
C	Normal	Cono de diamante	10	150	Negro	Fuera	Aceros duros. Con dureza superior a 100 Rockwell B ó 20 Rc.
D	Normal	Cono de diamante	10	100	Negro	Fuera	Aceros cementados.
E	Normal	Bola de 1/8"	10	100	Rojo	Dentro	Metales blandos, como antifricción y piezas fundidas.
F	Normal	Bola de 1/16"	10	60	Rojo	Dentro	Bronce recocido.
G	Normal	Bola de 1/16"	10	150	Rojo	Dentro	Bronce fosforoso y otros metales.
H	Normal	Bola de 1/8"	10	60	Rojo	Dentro	Metales blandos, con poca homogeneidad, fundición de hierro.
K	Normal	Bola de 1/8"	10	150	Rojo	Dentro	Metales duros, con poca homogeneidad, fundición de hierro.
L	Normal	Bola de 1/4"	10	60	Rojo	Dentro	Metales duros, con poca homogeneidad, fundición de hierro.
M	Normal	Bola de 1/4"	10	100	Rojo	Dentro	Metales duros, con poca homogeneidad, fundición de hierro.
P	Normal	Bola de 1/4"	10	150	Rojo	Dentro	Metales duros, con poca homogeneidad, fundición de hierro.
R	Normal	Bola de 1/2"	10	60	Rojo	Dentro	Metales muy blandos.
S	Normal	Bola de 1/2"	10	100	Rojo	Dentro	Metales muy blandos.
V	Normal	Bola de 1/2"	10	150	Rojo	Dentro	Metales muy blandos.
15-N	Superficial	Cono de diamante	3	15	Rojo	Dentro	Aceros nitrurados, cementados y de herramientas de gran dureza.
30-N	Superficial	Cono de diamante	3	30	Rojo	Dentro	Aceros nitrurados, cementados y de herramientas de gran dureza.
45-N	Superficial	Cono de diamante	3	45	Rojo	Dentro	Aceros nitrurados, cementados y de herramientas de gran dureza.
15-T	Superficial	Bola de 1/16"	3	15	Rojo	Dentro	Bronce, latón y acero blando.
30-T	Superficial	Bola de 1/16"	3	30	Rojo	Dentro	Bronce, latón y acero blando.
45-T	Superficial	Bola de 1/16"	3	45	Rojo	Dentro	Bronce, latón y acero blando.

ROCKWELL HARDNESS (HR)

Advantages

- Fast
- Test for all types of materials.
- Very precise ($\pm 0.002\text{mm} \Leftrightarrow \pm 1\text{unit}$)
- Operator errors have little effect.

Polymers:

As a consequence of the high ability for plastic deformation at room temperature, the load has to be applied for 15s.

Metals:

Time of application of the load [1-15]s depending on the melting T of the metal.

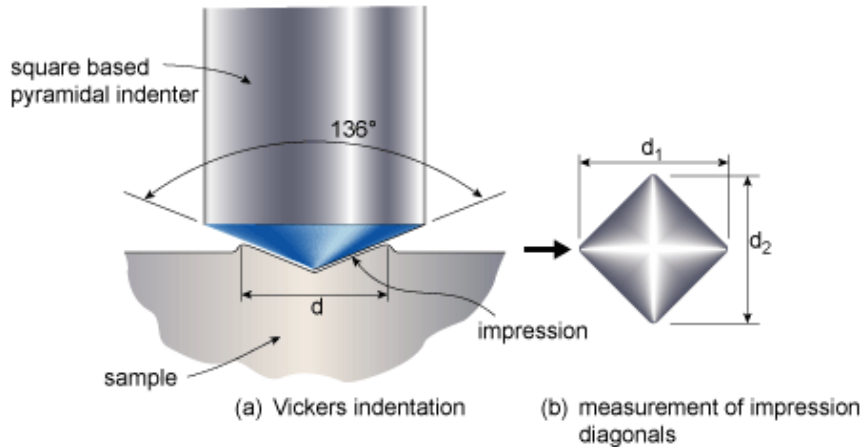
Specific scales for coatings

VICKERS HARDNESS (HV)

VICKERS HARDNESS (HV)

Indenter: diamond cut rectangular pyramid 136° between faces

Nomenclature: HV or VHN= Vickers Hardness



The value of hardness is a function of the indentation diagonal = f (area of the pyramid)

HV30 or HV15 or HV10.....HVkgf

$$HV \approx \frac{\text{Load}}{\text{Area indentation}} = \frac{2F \sin \frac{136^\circ}{2}}{d^2} = 1.854 \frac{F}{d^2}$$

F = Load in kgf [1 - 120]kg

d = Arithmetic mean between d_1 and d_2 in mm.

Time = 10-15 s

Measure: 345 HV30-25 (≈HVkgf-s)

Advantages of Vickers method

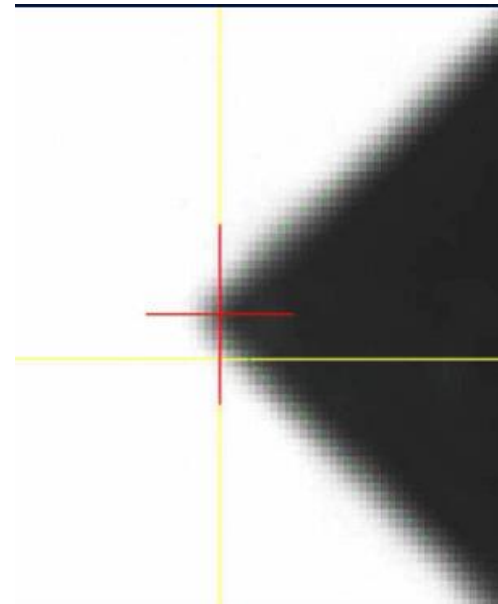
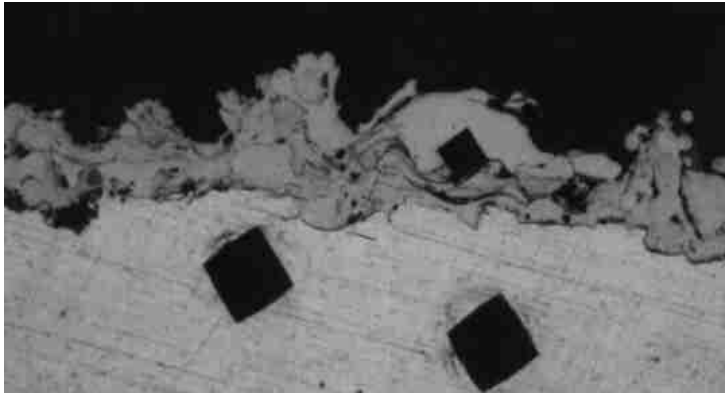
- Universal result, the value is independent of the load
- With the indenter we measure a wide range of materials
- Possibility to measure thin components ($\geq 0.05\text{mm}$ thickness)
- Can measure surface hardness
- More precise than Rockwell scale (1unit HRX = 32,5 unit HV)

VICKERS HARDNESS (HV)

Microhardness

Applied load does not exceed 1 kgf.

Measures hardness locally, in individual constituents of a microstructure.



Variability in the measurements due to error by the operator

HARDNESS TEST

Precautions for hardness tests

- Flat surface and $\perp$ axes to the application of the load
- Polished and clean from oxides, carbides, oils ,..... surface
- Distance between the edge of the component and the indentation $> 4 \cdot \varnothing$ (Brinell or Rockwell)
- Specimen thickness:
 - $> 2 \cdot \varnothing$ (Brinell or Rockwell)
 - > 1.5 times the dent diagonal (Vickers)

METALS

Material	HBW	HR*
Heat treated Al	20	
Hardened Al alloys	90	52 B
Fe cast	175	85 B
Bronze	100	60 B
Pb	4	
C Steel	150	10 C
Ti high purity	200	95 B

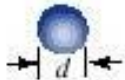
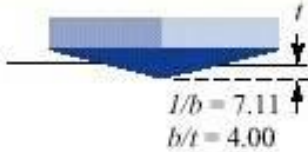
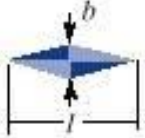
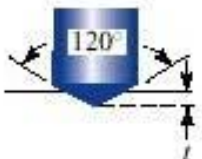
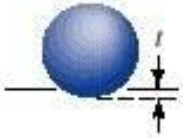
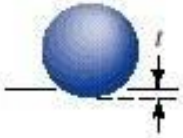
CERAMICS

Material	HV	HK, Knoop
WC-Co	2000	1400
Al ₂ O ₃	2200	1500
SiC	2600	1900
TiC	3200	2500
BN	6000	4000
Sintered Diamond	7000	5000
Natural Diamond	10000	8000

POLYMERS

Material	HBW
Nylon	12
LDPE	2
HDPE	4
PP	7
PS	20
PVC	10

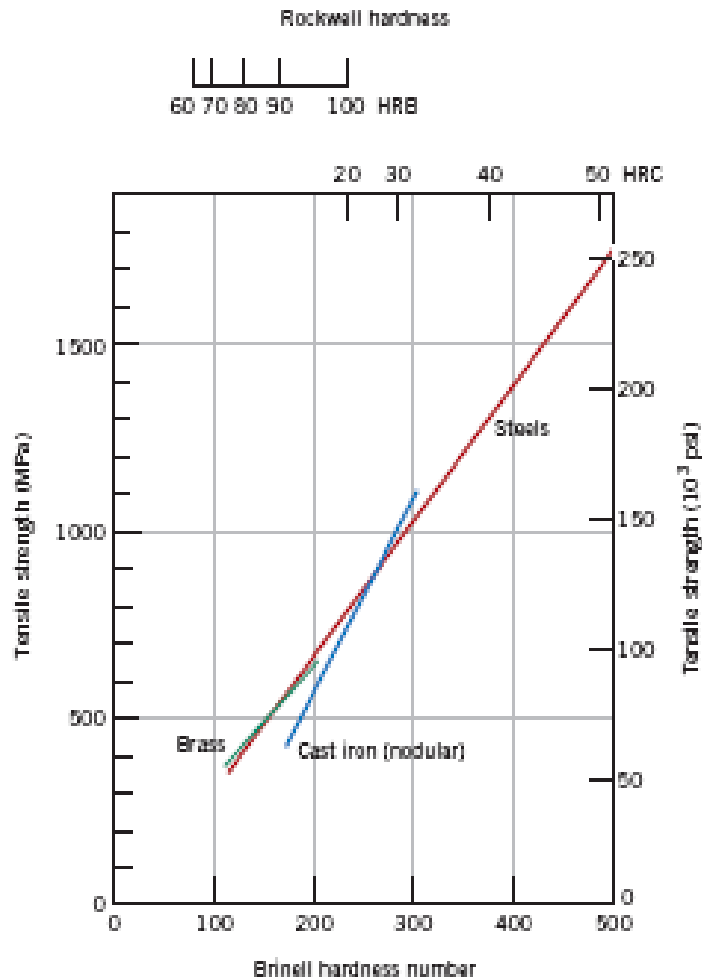
HARDNESS SCALE

Test	Indenter	Shape of Indentation		Load	Formula for hardness number
		Side view	Top view		
Brinell	10 mm sphere of steel or tungsten carbide			P	$\text{BHN} = \frac{2P}{\pi D \left[D - \sqrt{D^2 - d^2} \right]}$
Vickers	Diamond pyramid			P	$\text{VHN} = 1.854 \cdot P/d^2$
Knoop microhardness	Diamond pyramid			P	$\text{KHN} = 14.2 P/l^2$
Rockwell	A } C } D }			60 kg	$R_A =$
				150 kg	$R_C =$
				100 kg	$R_D =$
	B } F } G }			100 kg	$R_B =$
				60 kg	$R_F =$
				150 kg	$R_G =$
	E } H }			100 kg	$R_E =$
				60 kg	$R_H =$

source: H. W. Hayden, W. G. Moffatt, and J. Wulff, *The Structure and Properties of Materials*, Vol. 3: *Mechanical Behavior*, John Wiley & Sons, Inc., New York, 1965.

HARDNESS

Correlation between hardness, elastic limit and σ_{tension}



The tensile strength and hardness are indicators of plastic deformation → The relation = f (material)